CASCADIA ANALYTICS

Moderna, Inc.

Equity Research Note | MRNA | 21 OCTOBER 2026

Platform Rich, Business Poor

Moderna screens as a Sell with a $95.15 blended target, 61% below the current price.

Analyst: Justin Wen

REFERENCE PRICE	BLENDED TARGET	BLENDED UPSIDE	
$242.08	$95.15	-61%	SELL

Investment Thesis

Platform Rich, Business Poor. Real mRNA moat, $6.9B cash [10-Q Q2 26], flu vaccine approved. But revenue fell from $19.3B (2022) to $1.94B (2025) [8-K FY25], and the business still loses ~$3B a year. Binary Oncology Option. Intismeran Phase III melanoma data on Oct 24, 2026 is the swing factor. The 306% six-month run has priced in success; a miss unwinds fast toward the $101 consensus target [Cons]. Price Denies the DCF. At $242.08 [Mkt 17-Oct-26], the market implies 122% revenue CAGR. Our base case $89.18 means

- Platform Rich, Business Poor
- Binary Oncology Option
- Price Denies the DCF

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The latest print

Binary Oncology Option. Intismeran Phase III melanoma data on Oct 24, 2026 is the swing factor. The 306% six-month run has priced in success; a miss unwinds fast toward the $101 consensus target.

Price Denies the DCF. At $242.08, the market implies 122% revenue CAGR. Our base case $89.18 means -63% downside, and the bull case $182.44 does not even reach the current price.

$1.94B	$145M	-$2.82B	$6.9B	up to +10%	$3.39B
REVENUE (FY2025)	REVENUE (Q2 2026)	NET LOSS (FY2025)	CASH & INVESTMENTS (JUN 30, 2026)	2026 REVENUE GUIDE	2028 REVENUE CONSENSUS

Key metrics in detail

METRIC	LATEST	VS YEAR AGO	TREND
Revenue (FY2025)	$1.94B	-39.9%	Down
Revenue (Q2 2026)	$145M	+2.1%	Flat
Net Loss (FY2025)	-$2.82B	-$7.26/sh	Down
Cash & Investments (Jun 30, 2026)	$6.9B	$4.7-5.2B guide yr-end	Down
2026 Revenue Guide	up to +10%	off $1.94B base	Up
2028 Revenue Consensus	$3.39B	+75% vs 2025	Up

The moat

Intismeran personalized cancer vaccine with Merck; Phase III melanoma interim in H2 2026

The growth engine

Respiratory portfolio expansion: mNEXSPIKE rollout, FDA-approved flu vaccine, EU market opening 2027

Three scenarios make the valuation range explicit

Three scenarios, weighted 25% / 50% / 25% into the blended target.

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Moderna, Inc. valuation framework

SCENARIO	WEIGHT	2026-30 CAGR	2030 REVENUE	2030 EBIT MARGIN	TARGET	UPSIDE
Bear	25%	20.1%	$2.50B	15.0%	**$19.78**	-92%
Base	50%	51.6%	$8.00B	32.0%	**$89.18**	-63%
Bull	25%	67.0%	$13.00B	42.0%	**$182.44**	-25%

Bear: downside case at $19.78 (-92%).

Base: central case at $89.18 (-63%).

Bull: upside case at $182.44 (-25%).

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The probability-blended target implies 61% downside

A 25% Bear / 50% Base / 25% Bull weighting implies -61% from the $242.08 reference price.

$95.15
BLENDED TARGET

The valuation uses a multi-year unlevered free cash flow DCF. The discount rate is the live CAPM rate: the 10-year Treasury yield plus beta times the market-implied equity risk premium. Terminal value is a Gordon perpetuity on terminal-year free cash flow.

Mechanics are explicit. The bridge runs revenue to EBIT to cash taxes, then adds back D&A and subtracts capex and working-capital investment to reach unlevered free cash flow. Targets are divided by diluted shares outstanding.

Revenue base (2025E)	**$1.00B**
Valuation horizon	**2026-2030**
Live WACC	**8.75%**
WACC path (Year 1 to final year)	**8.75% to 7.97%**
Terminal growth	**2.0%**
Net cash	**$3.82B**
2030 FCF (Base)	**$2.63B**
10-year Treasury yield	**5.1%**
Beta	**1.09**
Equity risk premium	**3.4%**

SELL | $95.15 BLENDED TARGET | WEN

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The reverse DCF: what the market prices in

Holding the base-case FCF margin flat, the market-implied revenue CAGR is 129.1%. market prices ~122% 5-yr revenue CAGR — euphoric, far above Bull case (requires revenue $1B → $40B+ by 2030)

Reverse DCF sensitivity

Implied enterprise value and share price at each five-year revenue CAGR, with the base-case free cash flow margin held flat. The highlighted row sits nearest the market-implied CAGR.

CAGR	Implied EV	Implied price	CAGR	Implied EV	Implied price	CAGR	Implied EV	Implied price
-10%	$2.7B	$16.32	+10%	$5.6B	$23.53	+30%	$10.4B	$35.65
-9%	$2.8B	$16.59	+11%	$5.8B	$24.01	+31%	$10.7B	$36.41
-8%	$2.9B	$16.87	+12%	$6.0B	$24.50	+32%	$11.0B	$37.20
-7%	$3.0B	$17.16	+13%	$6.2B	$25.00	+33%	$11.3B	$38.00
-6%	$3.1B	$17.45	+14%	$6.4B	$25.51	+34%	$11.7B	$38.82
-5%	$3.3B	$17.76	+15%	$6.6B	$26.03	+35%	$12.0B	$39.66
-4%	$3.4B	$18.07	+16%	$6.8B	$26.57	+36%	$12.3B	$40.51
-3%	$3.5B	$18.39	+17%	$7.0B	$27.13	+37%	$12.7B	$41.38
-2%	$3.7B	$18.73	+18%	$7.2B	$27.69	+38%	$13.0B	$42.28
-1%	$3.8B	$19.07	+19%	$7.5B	$28.27	+39%	$13.4B	$43.19
+0%	$3.9B	$19.42	+20%	$7.7B	$28.87	+40%	$13.8B	$44.12
+1%	$4.1B	$19.78	+21%	$7.9B	$29.48	+41%	$14.2B	$45.07
+2%	$4.2B	$20.16	+22%	$8.2B	$30.10	+42%	$14.6B	$46.04
+3%	$4.4B	$20.54	+23%	$8.4B	$30.74	+43%	$14.9B	$47.03
+4%	$4.5B	$20.93	+24%	$8.7B	$31.39	+44%	$15.3B	$48.04
+5%	$4.7B	$21.34	+25%	$9.0B	$32.06	+45%	$15.8B	$49.07
+6%	$4.9B	$21.75	+26%	$9.2B	$32.75	+46%	$16.2B	$50.12
+7%	$5.0B	$22.18	+27%	$9.5B	$33.45	**+47%**	**$16.6B**	**$51.20**
+8%	$5.2B	$22.62	+28%	$9.8B	$34.16			
+9%	$5.4B	$23.07	+29%	$10.1B	$34.90			

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Balance sheet and share count reconciliation

Enterprise value converts to equity value after net debt, and targets divide by diluted shares.

SCENARIO	ENTERPRISE VALUE	EXIT EV/ FCF	TERMINAL GROWTH	PLUS: NET CASH	EQUITY VALUE	TARGET
Bear	$4.07B	—	2.0%	$3.82B	$7.89B	**$19.78**
Base	$31.76B	—	2.0%	$3.82B	$35.58B	**$89.18**
Bull	$68.97B	—	2.0%	$3.82B	$72.79B	**$182.44**

Fair value across scenarios

Model fair value by year under each scenario, on the same paths the ticker page chart draws.

SCENARIO	2026	2026	2027	2028	2029	2030
Bear	$21.49	$21.49	$27.14	$31.74	$35.36	$38.09
Base	$96.86	$96.86	$108.29	$119.63	$128.87	$136.17
Bull	$198.14	$198.14	$217.35	$236.78	$253.92	$267.23

Methodology note. The discount rate is the live U.S. 10-year Treasury yield plus beta times equity risk premium. Terminal growth reflects long-run nominal GDP. The $95.15 headline target applies 25% / 50% / 25% Bear / Base / Bull weights.

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The franchise deep dive

Intismeran personalized cancer vaccine with Merck; Phase III melanoma interim in H2 2026

Respiratory portfolio expansion: mNEXSPIKE rollout, FDA-approved flu vaccine, EU market opening 2027

Cost discipline: 2025 operating expenses cut $2.2B (30%), 2026 GAAP opex guided to ~$4.7B ex litigation

Revenue and margin trend

METRIC	FY22	FY23	FY24	FY25	TTM
Revenue	$19.26B	$6.85B	$3.24B	$1.94B	$2.23B
Net Loss	n/a	n/a	n/a	-$2.82B	-$3.20B
Cash & Investments	n/a	n/a	n/a	$8.1B	$6.9B

Valuation cross-check

P/S (trailing) 44x

P/B 9.7x

DCF Base $89.18, -63% downside

The market prices 24x consensus 2028 revenue for a business still losing billions. Our DCF says the price is not even supported by the bull case ($182.44).

BNTX, NVAX surged too

Consensus target $101, Hold

MRNA trades at the richest multiple among vaccine biotechs while 26 analysts rate it a Hold with targets 58% below the price.

$242.08 vs $22 Nov 2025 low

10x rally on no revenue growth

The entire move is multiple expansion on pipeline hope, not financials. Prior peaks collapsed 90%+.

What to monitor

24-Oct-2026. Intismeran Melanoma Data (Positive): Phase III interim readout with Merck; the single biggest swing factor for the pipeline premium.

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05-Nov-2026. Q3 2026 Earnings (Two-Sided): Consensus -$1.10 EPS; U.S. vaccination rate and fall rollout color will matter more than the print.

2026. Flu Vaccine Launch (Positive): mRNA-1010 FDA approved Sep 2026; first commercial season diversifies beyond COVID.

2026. Norovirus Phase III Readout (Two-Sided): Earlier interim missed early statistical criteria; additional cohort enrolled.

Financial summary: income statement

YEAR	REVENUE	GROSS PROFIT	OPERATING INCOME	EBITDA
2021	$18.5B	—	$13.3B	—
2022	$18.9B	$13.5B	$9.4B	$9.8B
2023	$6.8B	$2.1B	$-4.2B	$-3.6B
2024	$3.2B	$1.7B	$-3.9B	$-3.4B
2025	$1.9B	$1.1B	$-3.1B	$-2.5B

Balance sheet

YEAR	TOTAL ASSETS	TOTAL LIABILITIES	TOTAL EQUITY	CASH
2022	$25.9B	$6.7B	$19.1B	$3.2B
2023	$18.4B	$4.6B	$13.9B	$2.9B
2024	$14.1B	$3.2B	$10.9B	$1.9B
2025	$12.3B	$3.7B	$8.7B	$2.6B

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Cash flow

YEAR	OPERATING CF	CAPEX	FINANCING CF	FREE CF
2022	$5.0B	$-400M	$-3.4B	$4.6B
2023	$-3.1B	$-707M	$-1.4B	$-3.8B
2024	$-3.0B	$-1.1B	$56M	$-4.1B
2025	$-1.9B	$-202M	$593M	$-2.1B

Methodology note. The discount rate is the live U.S. 10-year Treasury yield plus beta times equity risk premium. Terminal growth reflects long-run nominal GDP. The $95.15 headline target applies 25% / 50% / 25% Bear / Base / Bull weights.

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Agency risk score 65 of 100: balanced, typical corporate alignment

The score follows the website's Executive Alignment & Agency Risk model: insider ownership up to 40 points, open-market Form 4 flow over the trailing six months up to 30, and pay-for-performance up to 30. Insiders hold 7.36% of shares outstanding. Form 4 filings over the trailing six months show $0 of open-market purchases against $55.38M of sales, a net flow of -$55.38M. One-year total shareholder return is +801.6% against CEO total pay of $6.14M (Mr. Stéphane Bancel M.B.A.), which the model scores as PERFORMANCE EARNED.

65

AGENCY RISK SCORE, OF 100

The three components

COMPONENT	SCORE	WHAT THE SITE SHOWS
Insider Ownership (Skin in the Game)	25 / 40	7.36% of shares held by insiders
Form 4 Flow (Conviction Flow)	14 / 30	Buys $0 (0) vs sales $55.38M (10), net -$55.38M, trailing six months
Pay-for-Performance	26 / 30	PERFORMANCE EARNED: Stock Outperformance Matches Comp Structure. 1Y TSR +801.6%; Mr. Stéphane Bancel M.B.A., CEO & Director, total pay $6.14M
Agency Risk Score	**65 / 100**	Regime: BALANCED / TYPICAL CORPORATE

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Recent Form 4 filings

DATE	FILER	RELATION	TYPE	SHARES	VALUE
2026-09-15	Hoge Stephen	President	SELL	80,575	$11.69M
2026-09-10	Mock James M	Chief Financial Officer	SELL	34,836	$4.69M
2026-09-01	Klinger Shannon Thyme	Officer	SELL	3,471	$485.8K
2026-08-05	Bancel Stephane J	Chief Executive Officer	SELL	499,246	$28.72M
2026-07-15	Hoge Stephen	President	SELL	53,336	$3.61M
2026-06-15	Hoge Stephen	President	SELL	53,336	$2.74M
2026-06-04	Klinger Shannon Thyme	Officer	SELL	3,471	$173.6K
2026-05-21	Afeyan Noubar Boghos	Director	SELL	9,263	$433.9K

SEC proxy audit, DEF 14A filed 2026-03-16

Alignment grade D. ABOVE-TARGET BONUS DESPITE REVENUE MISS. Stéphane Bancel, fiscal 2025 vs fiscal 2024; total compensation change year on year: N/A.

BASE SALARY	STOCK AWARDS	NON-EQUITY INCENTIVE	OTHER COMPENSATION
$1,687,200	$15,000,000	$4,302,360	$0

Looting flag: pay up sharply while shareholders lost money.

Annual cash incentive paid out at 170% of target ($4.30M) despite product revenue falling short of internal targets and three-year TSR ranking lowest among peers. The Compensation Committee justified the above-target multiplier through significant cash cost reductions of over $2 billion and pipeline milestone execution. While prior equity grants remain largely underwater, awarding heightened cash bonuses during a contracting commercial market reflects weak accountability to top-line performance and shareholder value.

Proxy audit source: SEC EDGAR DEF 14A, audited 2026-10-18 by the weekly proxy pipeline. Filing: https://www.sec.gov/Archives/edgar/data/1682852/000130817926000081/mrna-20260316.htm

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Methodology. This module reuses the website's Executive Alignment & Agency Risk model unchanged. Scores of 75 and above band as founder-mode, high conviction; 45 to 74 as balanced, typical corporate; below 45 as high agency risk. Inputs are the same feeds behind the site model (Yahoo Finance insider holdings, Form 4 transactions and executive compensation) plus the SEC DEF 14A CD&A audit from the weekly proxy pipeline. Computed 21 October 2026 for this note.

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The bear case

COVID Revenue Collapse

COVID revenue is in secular decline while the pipeline is years from meaningful revenue. Q2 product sales fell 18% YoY; U.S. vaccination rates are the key variable management cannot control.

Regulatory Risk

The FDA issued a refusal-to-file letter for the mRNA-1010 flu application on trial-design grounds despite prior protocol discussions. Regulatory unpredictability is now a proven risk to the platform.

Legal Costs and Dilution

A $950M Arbutus/Genevant litigation payment hit cash in July 2026, with ongoing royalties inflating cost of sales. The September $2.6B convertible adds 12.3M potential shares of dilution.

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The strengths

Operating Quality

Genuine mRNA platform moat with only BioNTech as a real peer, and a $6.9B cash fortress. But there is no recurring revenue engine: margins are -141%, ROIC deeply negative, and the business still burns billions a year.

Recent Trend

Revenue collapsed three years straight: -64%, -53%, -40%. Q2 2026 stabilized at +2% but off a $145M quarterly base. The real trend is cost discipline: opex cut $2.2B (30%) in 2025. Losses are narrowing, not because the business is growing.

Forward Growth

Credible but binary drivers: FDA-approved flu vaccine, mNEXSPIKE rollout, EU respiratory market opening 2027, Intismeran melanoma readout. Consensus only projects $3.39B revenue by 2028, 75% above 2025 but under half of 2023.

Valuation

At $242.08, market pays 44x trailing sales and 24x consensus 2028 revenue. Our DCF base is $89.18 (63% downside) and even the bull case $182.44 sits below the price. Reverse DCF implies 122% CAGR, pure perfection pricing. Consensus target $

Cascadia scorecard

Composite score 42: Unattractive. Confidence Medium, as of 17-Oct-2026.

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PILLAR	RATIONALE	SCORE
Operating Quality	Genuine mRNA platform moat with only BioNTech as a real peer, and a $6.9B cash fortress. But there is no recurring revenue engine: margins are -141%, ROIC deeply negative, and the business still burns billions a year.	55
Recent Trend	Revenue collapsed three years straight: -64%, -53%, -40%. Q2 2026 stabilized at +2% but off a $145M quarterly base. The real trend is cost discipline: opex cut $2.2B (30%) in 2025. Losses are narrowing, not because the business is growing.	40
Forward Growth	Credible but binary drivers: FDA-approved flu vaccine, mNEXSPIKE rollout, EU respiratory market opening 2027, Intismeran melanoma readout. Consensus only projects $3.39B revenue by 2028, 75% above 2025 but under half of 2023.	55
Valuation	At $242.08, market pays 44x trailing sales and 24x consensus 2028 revenue. Our DCF base is $89.18 (63% downside) and even the bull case $182.44 sits below the price. Reverse DCF implies 122% CAGR, pure perfection pricing. Consensus target $101 is -58% away.	25
Catalysts & Risk	Real near-term catalysts: Intismeran Oct 24 presentation, Q3 earnings Nov 5, flu launch. But risk skew is sharply negative at a 52-week high after a 306% six-month run. $950M Arbutus payment, $2.6B convert dilution, and FDA refusal-to-file history add overhang.	40

Investment view

The Oct 24, 2026 Intismeran melanoma readout is the swing factor. A clean positive validates the oncology pipeline and the platform premium; a miss or statistical ambiguity unwinds the 306% rally toward the $101 consensus target and our $89.18 DCF base.

Investment view: Sell. The $95.15 blended target implies -61%, balancing the $19.78 / $89.18 / $182.44 Bear / Base / Bull outcomes.

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